MIT GLOBAL SCALE NETWORK

A power company operates three power generation plants. One is a wind plant, and the other two consume a combination of Fuel 1 and Fuel 2, emitting carbon dioxide in the process. In addition, all three plants require maintenance. The amount of fuel consumed (in Mg), maintenance required (in person-hours), carbon dioxide (CO2) emitted (in Mg), and power generated (in MWh) per day of operation is as follows:

Plant	Maintenance Required	Fuel 1 Required	Fuel 2 Required	CO_2 Emitted	Power Produced
1	20	0	0	0	20
2	13	10	15	12	32
3	18	30	40	29	40

Each MWh of power can be sold at £121 and there is no limit on the amount that can be sold. Over its next planning period, the company has 230 person-hours for maintenance, 75 Mg of Fuel 1, and 90 Mg of Fuel 2 available.

(a) **60 points:** Due to environmental regulations, they cannot emit more than 200Mg of CO2 in this period. The company wants to know how to operate its plants to generate as much revenue as possible (you may assume that there is no limit on the number of days a plant can operate in this period). Give a linear program (only the formulation, you do not need to solve it) that models this problem and clearly state what each of your variables is meant to represent.

(b) **40 points:** Suppose now that the company can emit more than 200Mg of CO2, but now loses $55 of revenue for each Mg emitted after the first 200Mg because it must purchase "CO2 credits". The other resource constraints remain as stated. The company now wants to know how to operate its plants to generate as much revenue as possible (you may assume that there is no limit on the number of days that a plant can operate in this period). Give a linear program that models this problem (again, you do not need to solve this program).

Solution (part a:60 points)

Decision variables:

- x_i: number of days plant i is operated (i=1,2,3)

Parameters:

- m_i: maintenance required per day of operation of plant i *(20, 13, 18 for i=1,2,3 respectively)*
- $fone_i$: fuel 1 required (in Mg) per day of operation of plant i *(0, 10, 30 for i=1,2,3 respectively)*
- $ftwo_i$: fuel 2 required (in Mg) per day of operation of plant i *(0,15,40 for i=1,2,3 respectively)*
- $co2_i$: CO2 emitted (in Mg) per day of operation of plant i *(0,12,29 for i=1,2,3 respectively)*
- p_i: power produced (MWh) per day of operation of plant i *(20,32,40 for i=1,2,3 respectively)*
- s: sales price of one MWh (£121)

The formulation will be as follows:

$$\text{Max } z = \sum_{i=1}^{3} s \times p_i \times x_i$$

s.t.

$$\sum_{i=1}^{3} m_i x_i \leq 230$$

$$\sum_{i=1}^{3} fone_i x_i \leq 75$$

$$\sum_{i=1}^{3} ftwo_i x_i \leq 90$$

$$\sum_{i=1}^{3} co2_i x_i \leq 200$$

$$x_i \geq 0, \forall i = 1,2,3$$

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Solution (part b:40 points)

The decision variables, parameters, and the new formulation will be as follows:
Decision variables:

- x_i: number of days plant *i* is operated (*i*=1,2,3)
- *TCO2*: total amount of CO2 (in Mg) to be accounted for the carbon credits – *this is not the total CO2 emissions*

Parameters:

- m_i: maintenance required per day of operation of plant *i (20, 13, 18 for i=1,2,3 respectively)*
- $fone_i$: fuel 1 required (in Mg) per day of operation of plant *i (0, 10, 30 for i=1,2,3 respectively)*
- $ftwo_i$: fuel 2 required (in Mg) per day of operation of plant *i (0,15,40 for i=1,2,3 respectively)*
- $co2_i$: CO2 emitted (in Mg) per day of operation of plant *i (0,12,29 for i=1,2,3 respectively)*
- p_i: power produced (MWh) per day of operation of plant *i (20,32,40 for i=1,2,3 respectively)*
- *s*: sales price of one MWh (£121)
- *cf*: CO2 credit fee for each Mg of emission beyond 200 Mgs (£55)

The formulation will be as follows:

$$\text{Max } z = \sum_{i=1}^{3} s \times p_i \times x_i - cf \times (TCO2 - 200)$$

s.t.

$$\sum_{i=1}^{3} m_i x_i \leq 230$$

$$\sum_{i=1}^{3} fone_i x_i \leq 75$$

$$\sum_{i=1}^{3} ftwo_i x_i \leq 90$$

$$\sum_{i=1}^{3} co2_i x_i \leq TCO2$$

$$200 \leq TCO2$$

$$x_i \geq 0, \forall i = 1,2,3$$

$$TCO2 \geq 0$$